Predicting Cross-Sectional Relative Returns in the S&P 500: A Feature-Engineering and Walk-Forward Modeling Case Study

Quantitative Research Case Study

1 Problem and Approach

The task is to predict *relative* future stock returns inside the S&P 500: not whether the market goes up, but which stocks will out- or under-perform their peers. This framing is deliberate. Index-level direction is dominated by macro factors that daily OHLCV data cannot forecast, whereas the *cross-section* of returns carries persistent, tradable structure (momentum, reversal, liquidity, volatility, and sector-rotation effects) that is learnable from price and volume alone.

We therefore predict each stock's forward return *in excess of its market and sector*, first test validation-selected model ensembles, and then use the same machinery to build a narrower momentum-plus-residual overlay. The primary trading evaluation is a cost-adjusted, sector-neutral decile long–short portfolio: within each sector, long the top-scored stocks and short the bottom-scored stocks, then combine sectors into a dollar-neutral book. The entire pipeline is built to be honest about look-ahead leakage, overlapping-window inflation, transaction costs, and survivorship bias, because in this problem the easiest way to produce a spectacular-looking result is to make a methodological mistake.

Headline design. Curated technical/liquidity/cross-sectional features → sector- and market-relative forward-return targets → a single-factor momentum baseline → compact Ridge/XGBoost models trained on factor-neutral residual labels → a validation-selected momentum-plus-residual overlay → rank IC / ICIR, DSR/PBO, style-neutral attribution, capacity, and cost-aware portfolio backtests. The headline result is no longer a graph or MLP claim: it is a 30-trading-day sector-neutral residual overlay that improves hold-out Sharpe and drawdown versus standalone momentum, while still failing the strict full-grid multiple-testing hurdle.

2 Data

We use only two files: `sp500_stocks.csv` (daily OHLCV per ticker) and `sp500_companies.csv` (sector, sub-industry, headquarters, founding year, and index `date_added`). No fundamentals, analyst, macro, or external listing data are used. The cleaned panel covers 503 symbols from 2000-01 to 2026-06 (2.926M stock-day rows).

Price adjustment (data quality). The raw file exposes only a `close` column and no separate `adj_close`. We therefore audited major split events and verified that this `close` behaves as an adjusted close price: around NVDA's 2024-06-10 10:1 split (120.68 → 121.58) and AAPL's 2020-08-31 4:1 split (121.06 → 125.17) there is no split-day gap, consistent with a `yfinance auto_adjust=True` export. In this study, `close` is therefore treated as the adjusted close used for all returns, momentum, volatility, and forward-return targets. Residual uncertainty about dividend adjustment is treated as a minor data-quality caveat.

Raw cleaning. The CSV-to-Parquet preparation step normalizes ticker strings, enforces one row per `symbol,date`, drops rows with missing core OHLCV fields, non-positive prices or volume, and impossible OHLC relationships (`high` below open/low/close or `low` above open/high/close). The raw stock file has no core missing values or duplicate stock-days, but it does contain 4,413 zero-volume rows and one OHLC inversion; these 4,414 rows are removed before feature construction. Large but plausible moves are audited rather than deleted: 44 stock-days have absolute one-day returns above 50%, while none exceed the 300% hard error threshold. Company metadata are also normalized and deduplicated; all 503 tickers have sector and sub-industry labels.

Survivorship bias. The constituent list is effectively the *current* membership applied to history, so firms that went bankrupt, were delisted, or were acquired (Enron, Lehman, Bear Stearns, WaMu, Sears, Kodak, Yahoo, Celgene, Xilinx, . . .) are absent. This biases backtests optimistically and is discussed quantitatively in §6. We do not claim to correct it; we bound its impact and avoid designs that amplify it.

3 Feature Engineering

Each row is one stock-day observation, and every feature uses only information available at or before that date (all rolling windows are backward-looking with an explicit `shift`). We build ≈500 features in economically motivated groups:

- **Price / momentum / reversal:** multi-horizon returns (1–252d), log returns, skip-window momentum, momentum acceleration, short-term reversal, intraday/overnight decomposition, close-in-bar location. *Captures short-term reversal and medium-term momentum.*
- **Volatility / tail / drawdown:** rolling, upside/downside, Parkinson and Garman–Klass volatility, ATR, skew/kurtosis, drawdown from recent highs. *Controls for risk state and crash sensitivity.*
- **Trend / technical:** SMA/EMA gaps and slopes, MACD, RSI, Bollinger z-score/bandwidth/%b, channel position, breakout distance, Williams %R, CCI. *Encodes overbought/oversold and trend regimes.*
- **Liquidity / volume:** log dollar volume, volume z-scores and momentum, Amihud illiquidity, return-per-dollar-volume. *Captures attention, liquidity shocks, and transaction-cost pressure.*
- **Market regime:** equal-weight market return, breadth, cross-sectional dispersion, market volatility. *Relative-return predictability is regime-dependent.*
- **Statistical linkage / residual risk:** rolling market beta, market correlation, idiosyncratic return, and idiosyncratic volatility. *Captures stock-level residual behavior after market exposure is removed.*
- **Sector / sub-industry and peer / style / geography:** group returns, breadth, dispersion, leave-one-out peer means, headquarters-region peers, and style buckets. *Separates stock-specific alpha from group movement.*
- **Graph relation features:** two graph feature families are kept separate. The original `fixed_graph` variant uses deterministic same-sector/sub-industry, style-nearest-neighbor, and rolling-correlation graph relation embeddings. The `supervised_graph` variant uses out-of-fold supervised GAT embeddings and scores. *This separates static peer-structure features from supervised relation-aware graph learning.*
- **Cross-sectional transforms:** date-wise z-scores/ranks plus sector- and sub-industry-neutral versions of the key signals. *Because the target is relative, cross-sectional normalization is usually more informative than raw levels.*

Feature audit and sanity check. The rebuilt feature manifest contains 525 non-target columns and 40 target columns. The large count is not 525 independent ideas; it comes from expanding a smaller set of economic signals across lookback windows, related definitions, and neutralization transforms. For example, the cross-sectional layer alone contributes 168 columns by applying date-wise, sector-wise, and sub-industry-wise z-scores and ranks to selected base signals. The default walk-forward models do not use the full catalog: they use a curated 61-column `core` set covering momentum/reversal, volatility, technical indicators, liquidity, market/industry context, and key cross-sectional ranks. The executable grid treats graph inclusion as a feature variant: `tabular` uses those 61 core features, while `fixed_graph` (called `graph` in some lower-level pipeline outputs) appends 19 deterministic graph columns (`graph_emb_0`–`graph_emb_15` plus relation-fusion weights for sector, style kNN, and rolling correlation). The three relation weights (`graph_rel_weight_sector`, `graph_rel_weight_style_knn`, and `graph_rel_weight_rolling_corr`) are normalized across relations and therefore sum to about one up to floating-point error; for each stock-date they indicate whether the graph relation embedding relies more on sector peers, style-nearest peers, or rolling-correlation peers. Separately, `supervised_graph` appends target-matched out-of-fold GAT score, embedding, and relation-weight columns, while `graph_supervised` is the combined fixed-graph plus supervised-GAT variant. All variants use the same targets, folds, models, costs, purge/embargo, and hold-out protocol.

We explicitly verified that the expected factor families are present: 12-minus-1-month momentum (`mom_252d_skip_21d`), short-term reversal (`rev_*`), volatility/range estimators, Bollinger features, `MACD`, RSI-style oscillators, Amihud illiquidity, and date/sector/sub-industry z-score and rank transforms. The short-cycle feature block also includes the signals most likely to be orthogonal to 12–1 momentum: 1–5 day reversal, Amihud liquidity/impact, volume shocks, market breadth, cross-sectional dispersion, and rolling idiosyncratic return after market-beta removal. As a usefulness check, we ran a single-factor daily rank-IC scan on the cleaned 2008–2026 panel using the same T+1 label shift, bottom-10% liquidity filter, and 1%/99% within-date winsorization as the modeling pipeline. Table 1 shows the largest effects in the core set. The magnitudes are small, as expected for daily cross-sectional equity alpha, but the signs are economically coherent: short-term overextension tends to reverse, 12–1 momentum is positive, and liquidity/impact variables contain weak but measurable information.

Prediction target. For horizon h we define the forward total return $r_{i,t}^{(h)}$ and subtract a contemporaneous group mean to obtain sector/market-relative targets, e.g. $\text{target\_excess\_sector\_fwd\_}h\text{d}_{i,t} = r_{i,t}^{(h)} - \bar{r}_{\text{sector}(i),t}^{(h)}$. Targets are forward-looking and are strictly excluded from the feature matrix. The target generator also writes a date-wise percentile-rank target, `target_rank_fwd_hd`, so short-horizon experiments can use volatility-normalized, ranked, or winsorized labels

Table 1: Single-factor sanity check on core features, target `target_excess_sector_fwd_5d`. Mean IC is the average daily Spearman rank correlation after the production cleaning/evaluation controls.

Feature	Mean IC	Interpretation
`sector_rank_sma_gap_20d`	-0.0131	within-sector short-term reversal
`ret_5d`	-0.0125	recent winners partially mean-revert
`amihud_20d`	+0.0123	liquidity / price-impact information
`log_dollar_volume`	-0.0123	liquidity/size relation in relative returns
`mom_252d_skip_21d`	+0.0102	12–1 month momentum survives weakly
`rsi_14d`	-0.0099	overbought names tend to cool off

when raw returns are too noisy. The implemented horizon grid covers $h \in \{1, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90\}$. The original broad search emphasized 5d/10d, but the final Route B rerun uses the 30d sector-relative horizon because the shorter horizons did not deliver reliable net economics after costs.

Execution timing. The raw target columns are generated on each feature date, but the model is evaluated with a one-day implementation lag. A feature row at date t is interpreted as information available after the t close; the signal is therefore tradeable only from t+1. In code, the effective training label is the raw target shifted one trading day forward, so a row dated t is matched to the forward return window that starts from the next trading day. This is the convention used for both rank-IC evaluation and portfolio PnL.

4 Methodology

4.1 Models

We compare, from simple to flexible: (i) a **single-factor momentum** baseline (rank by 12-minus-1-month momentum); (ii) **Ridge** regression (linear, strong regularization); (iii) **LightGBM** and (iv) **XGBoost** gradient-boosted trees (non-linear, interaction-aware); and (v) a small **PyTorch MLP** (two hidden layers, dropout, weight decay, early stopping on validation). The MLP is kept deliberately small and is *not* tuned until it wins; §5 reports its honest standing — competitive on in-sample validation IC, but not stable enough to headline once the untouched hold-out, turnover, and momentum benchmark are applied.

Momentum-residual overlay. Because the single-factor momentum baseline is strong, the implemented post-training evaluator does not force ML to replace momentum. For each date, both the baseline score and model score are converted to cross-sectional z-scores, the model score is residualized on the momentum score, and the final overlay signal is

$$s_{i,t} = z(\text{momentum}_{i,t}) + \lambda z(\widehat{\varepsilon}_{i,t}),$$

where $\widehat{\varepsilon}_{i,t}$ is the model component orthogonal to momentum within that day's cross-section. The overlay weight is selected from a small pre-registered grid on validation net portfolio metrics, with λ=0 reproducing the momentum baseline. The final Route B run selects λ=0.2 on validation Sharpe; the test split is not used to tune the amount of residual ML alpha.

Route B residual-target training. After the broad model/graph search failed to produce a robust replacement for momentum, I reran the main experiment with a narrower design: each effective forward label is residualized cross-sectionally by date against sector dummies and OHLCV style proxies (liquidity, volatility, beta/idiosyncratic volatility, short-term reversal, 12–1 momentum, overnight/intraday return, volume shock, breadth and dispersion). Ridge/XGBoost then train on this residual target. The final tradable signal is still evaluated on raw forward returns and is combined with the momentum baseline as a small residual sleeve rather than a replacement.

Graph feature variants. Graph features are predictive inputs, not separate trading rules. The `fixed_graph` variant is the original deterministic graph embedding: it builds sector, style-nearest-neighbor, and rolling-correlation relations from information available at the feature date and feeds the resulting `graph_emb_*` and relation-weight columns into the same tabular models. The `supervised_graph` variant is the GAT extension: a supervised GAT is trained on the same time splits, then emits an out-of-fold `supervised_graph_score`, relation weights, and low-dimensional embeddings. The optional `graph_supervised` variant combines both feature families. All graph variants are evaluated under the same purge, embargo, execution-lag, validation, cost, and portfolio rules as the tabular variant.

Table 2: Graph feature variants used in the model grid. The table separates deterministic graph embeddings from supervised GAT embeddings before any portfolio construction is applied.

Variant	Construction and label use	Added features and interpretation
`fixed_graph`	Deterministic sector, style-nearest-neighbor, and rolling-correlation peer relations; no target labels are used.	Adds `graph_emb_*` and fixed relation weights; tests whether static peer structure adds information to tabular models.
`supervised_graph` / GAT	Supervised relation-aware GAT trained on the same time splits; uses training labels only and is out-of-fold for scored rows.	Adds `supervised_graph_score`, GAT embeddings, and learned relation weights; tests whether target-aware graph representation helps ranking.
`graph_supervised`	Combination of deterministic fixed graph features and supervised GAT features; partly label-trained through the GAT component.	Adds both feature families; tests whether fixed peer structure and supervised GAT embeddings are complementary, at higher complexity and overfitting risk.

4.2 Validation-selected rank ensemble

The ensemble is the final *prediction layer*, not a portfolio construction rule. For each base model m and date t, raw predictions are converted into a cross-sectional percentile rank $q_{i,t}^{(m)} = \mathrm{rank}_t(\hat{y}_{i,t}^{(m)})$. The final score is then an average of member ranks,

$$\mathrm{final\_score}_{i,t} = \sum_{m \in \mathcal{M}} w_m q_{i,t}^{(m)},$$

with either equal weights or non-negative weights proportional to validation rank ICIR. Member selection uses validation rank ICIR and stability gates rather than validation mean IC; the post-validation test split is never used to select models, tune hyperparameters, choose ensemble members, or choose weights.

We compare five ensemble definitions:

- `ensemble_all_val_selected`: the current full ensemble, using the validation-selected best run for each model and feature variant.
- `ensemble_no_mlp`: the same rule after excluding all MLP members.
- `ensemble_tree_only`: Ridge, LightGBM, and XGBoost members only.
- `ensemble_supervised_graph_tree`: only supervised-graph Ridge, LightGBM, and XGBoost members.
- `ensemble_stability_filtered`: only members with positive validation rank IC, enough positive validation years, and non-negative available validation-regime rank IC.

This is also why the robustness tables report `ensemble_no_mlp` and `ensemble_tree_only`: the triage runs show that MLPs can look good on validation IC but are not stable enough to headline unless they pass the same walk-forward and regime gates.

4.3 Leakage controls

Every result below is produced under the same defenses:

- **One-day execution lag:** scores computed on day t are traded at $t{+}1$, so no same-bar information is used.
- **Label purge:** training rows whose forward label window crosses a split boundary are dropped, on both the training and the evaluation side.
- **Horizon embargo:** the validation/test window starts an h-day embargo after training ends, removing label overlap at the seam.
- **Train-only fitting:** imputation medians, scaling statistics, and the model are fit on training rows only.
- **Universe & winsorization:** before splitting, each date's cross-section drops the bottom 10% by dollar volume and clips features at the 1%/99% within-date quantiles, so illiquid names and outliers do not drive the fit or an untradeable backtest.

4.4 Avoiding overlapping-window inflation

A subtle but decisive issue: if an h-day forward return is booked every day, consecutive observations overlap by $h{-}1$ days, their IC series is strongly autocorrelated, and a naive ICIR $= \sqrt{252} \cdot \overline{\mathrm{IC}}/\sigma(\mathrm{IC})$ together with an overlap-booked Sharpe explode with h — a pure artifact, not alpha. We fix this in two ways: (1) the portfolio rebalances every h days so booked returns are *non-overlapping*; and (2) ICIR is annualized from a non-overlapping IC subsample (one observation per h days, scaled by $\sqrt{252/h}$). We retain the naive value as `rank_ic_ir_raw` only to document the inflation (Fig. 3). A sanity gate flags any run with Sharpe>3 or ICIR>5 as suspected leakage/overlap.

4.5 Validation: walk-forward selection + hold-out

A single static split can land in a lucky regime. The single-model horizon comparison therefore uses **expanding yearly walk-forward**: train on all data up to year Y, validate on year $Y+1$ (with purge + embargo), then roll forward one year and expand the training window. Model selection uses *walk-forward validation ICIR* and fold stability, not validation mean IC. The fixed 61-feature core is used instead of the full 525-column catalog for the same reason: it retains the economically interpretable, stable feature families while avoiding a high-variance short-horizon fit. The larger supervised-graph ensemble grid uses the same leakage controls but a fixed research split for tractability: train through 2018, validate through 2021, and evaluate the post-validation 2022–2026 test split once. In both cases, test results never participate in feature, model, hyperparameter, ensemble-member, or weight selection.

4.6 Portfolio and metrics

On each rebalance date we sort stocks by the reported score, either the validation-selected ensemble from §4.2 or the final Route B momentum-plus-residual overlay. Portfolio construction is deliberately written as a ladder from simple to stringent:

- **Long-only top bucket:** hold the top decile, either equal-weighted or score-weighted, with no short book. This is closest to a constrained long-only product and is evaluated with annualized return, volatility, Sharpe, max drawdown, turnover, and cost-adjusted return.
- **Plain long–short decile:** long the top 10%, short the bottom 10%, with equal dollar capital on both sides. This is the standard alpha ranking test corresponding to rank IC.
- **Sector-neutral long–short decile:** rank stocks within each sector, build top-minus-bottom decile books inside usable sectors, and average sector books into the total portfolio. This is the primary portfolio in the reported ensemble backtest because the target itself is sector-relative.
- **Market/sector beta-neutral portfolio:** an institutional-style robustness extension that constrains portfolio beta and sector exposures near zero in addition to dollar neutrality. It is stricter than the decile rule, but requires an optimizer and typically reduces raw return.
- **Cost-aware / turnover-controlled variants:** deduct explicit transaction costs, rebalance every 5/10/20 trading days or every target horizon, smooth scores with EWMA, and optionally restrict rebalancing to material signal changes through a no-trade band. The residual-turnover evaluator also sweeps no-trade bands of 0/5/10/15% of book turnover, which targets the gap between model turnover ($\sim$0.55–0.70 per rebalance) and the slower momentum benchmark ($\sim$0.16).

The empirical tables below use the sector-neutral long–short decile as the primary portfolio, rebalanced every h days with a 5 bps per-side transaction cost applied to realized turnover. We report rank IC, overlap-adjusted ICIR, decile spread, annualized net return, volatility, net Sharpe, max drawdown, and turnover.

5 Results

Validation selection. Selecting by across-fold mean validation ICIR, the small MLP is the strongest model at the 1-, 5-, and 30-day horizons (e.g. sector-relative 5d ICIR 1.15, market-relative 0.99) and LightGBM at 20 days (market-relative 0.99); Ridge and XGBoost trail (Fig. 2). No single model dominates every horizon and the boosted trees do *not* uniformly win — the MLP is genuinely competitive on the in-sample IC metric, so we do not pretend trees are categorically better.

Ensemble and graph comparison. The experiment runner supports three graph variants: deterministic fixed graph, supervised GAT, and their combined form. In code these are `fixed_graph`, `supervised_graph`, and `graph_supervised`. The archived ensemble metrics used for Table 3 contain the tabular and all-relation supervised-GAT runs for Ridge, LightGBM, XGBoost, and MLP, then construct the five validation-selected rank ensembles in §4.2. The fixed graph embedding is therefore documented as a model variant, but not included in this diagnostic table unless its corresponding archived metrics are present. Table 3 reports the sector-relative 5-day ensemble diagnostic on the test split using the primary sector-neutral long–short decile portfolio, net of 5 bps per-side cost. Excluding MLP improves the post-validation rank IC and net Sharpe relative to the full ensemble, and the supervised-graph tree ensemble is the strongest ranker among these 5-day ensembles; however, turnover remains high and economic performance is only weakly positive after costs. In this grid, `ensemble_no_mlp` and `ensemble_tree_only` coincide because the non-MLP model set is exactly Ridge, LightGBM, and XGBoost.

Table 3: 5-day sector-relative ensemble comparison on the 2022–2026 test split, using the primary sector-neutral long–short decile portfolio net of 5 bps per-side cost.

Ensemble	Weight	N	Rank IC	ICIR	Net Sharpe	Ann. Ret	Turnover
all val-selected	equal	8	0.0139	0.677	-0.34	-2.0%	0.68
all val-selected	ICIR	8	0.0126	0.608	-0.36	-2.1%	0.69
no MLP	equal	6	0.0168	0.807	-0.12	-0.7%	0.68
no MLP	ICIR	6	0.0168	0.801	0.04	0.3%	0.69
tree only	equal	6	0.0168	0.807	-0.12	-0.7%	0.68
tree only	ICIR	6	0.0168	0.801	0.04	0.3%	0.69
supervised-graph tree	equal	3	**0.0173**	**0.852**	-0.11	-0.7%	0.69
supervised-graph tree	ICIR	3	0.0171	0.835	**0.05**	**0.3%**	0.69
stability-filtered	equal	3	0.0061	0.208	-0.00	-0.0%	**0.66**
stability-filtered	ICIR	3	0.0051	0.210	-0.10	-0.6%	**0.66**

The decisive test: the untouched hold-out. The validation ranking does not survive out of sample. On the 2022–2026 hold-out the *single-factor momentum baseline* (rank by 12-minus-1-month momentum, no fitting) beats every trained multivariate model at the original 5-day benchmark horizon — higher rank IC (0.024 vs ≤ 0.013), higher ICIR (1.06 vs ≤ 0.73), higher net Sharpe (1.05 vs ≤ 0.53), and roughly one-quarter the turnover (0.16 vs ~ 0.59); the same ordering holds at 20 days (Table 9). The models' validation-IC edge is therefore largely overfit: a robust, low-turnover factor is hard to beat after costs. This is the central, deliberately humbling finding of the study.

Route B 30-day residual overlay. The final positive result comes from the narrower residual-target rerun, not from the graph/MLP zoo. XGBoost is trained on 30-day sector-relative labels after daily ridge residualization against sector and OHLCV style proxies. The tradable score is then $z(\text{momentum}) + 0.2z(\text{residual XGBoost})$, with $\lambda = 0.2$ selected on validation Sharpe. Table 4 compares this Route B overlay with standalone momentum on the 2022–2026 hold-out. The overlay only slightly improves raw Rank IC, but it materially improves the economic profile: Sharpe increases from 0.74 to 0.97 and max drawdown falls from -13.8% to -7.0%. Turnover rises, so capacity remains part of the risk budget.

Table 4: Route B 30-day sector-relative hold-out comparison, net of 5 bps/side cost. The model is not a replacement for momentum; it is a small residual sleeve added to it.

Config	Rank IC	ICIR	Net Sharpe	Ann. Ret	Max DD	Turnover
Momentum baseline	0.0282	0.698	0.740	7.78%	-13.83%	**0.401**
Route B overlay, $\lambda = 0.2$	**0.0283**	0.768	**0.970**	**8.08%**	**-6.96%**	0.492
Route B overlay, $\lambda = 0.3$ diagnostic	0.0280	**0.806**	1.201	9.98%	-6.10%	0.521

Earlier 10-day overlay diagnostic. Before the Route B residual-target rerun, the best-looking result was an incremental 10-day overlay on top of momentum. Table 5 is kept as a diagnostic because it motivated the final design: do not replace momentum; add only a controlled residual sleeve. It is no longer the headline result because the later P0 audit shows it does not survive multiple-testing and proxy factor-attribution checks.

Table 5: Earlier 10-day sector-relative residual overlay on the 2022–2026 hold-out. This diagnostic motivated Route B but is not the final headline.

Signal	Rank IC	ICIR	Net Sharpe	Ann. Ret	Max DD	Turnover
Momentum baseline	0.0271	0.812	0.746	7.05%	-14.59%	0.278
Residual overlay, validation-selected	0.0292	0.912	0.995	8.29%	-12.26%	0.368
Residual overlay, $\lambda = 1$ diagnostic	0.0332	1.205	1.297	9.44%	-7.60%	0.509

Graph relation ablation. The graph component was also tested as a proper relation ablation rather than accepted because a graph model looked good in isolation. The completed 120-candidate tree grid compares real relation families (sector, style-nearest-neighbor, rolling correlation, and all relations) against random-edge and no-edge controls for 5d/10d sector- and market-relative targets. Table 6 reports the best real relation and the strongest placebo/control in each target block. The result is mostly negative: real relations do not consistently beat random or no-edge controls, so graph learning is treated as an experimental feature family, not as a headline source of alpha.

Table 6: Graph relation ablation summary. ICIR is the overlap-adjusted hold-out rank-IC information ratio for the best tree/Ridge candidate in each block.

Target	Best real relation	ICIR	Best control	ICIR	Read-through
Sector 5d	style kNN / XGBoost	0.765	random / XGBoost	0.898	Placebo control wins; no graph-edge claim.
Market 5d	all relations / Ridge	0.505	random / Ridge	0.523	Placebo control wins on ICIR.
Sector 10d	style kNN / LightGBM	0.812	random / XGBoost	0.932	Real edges do not beat the random control.
Market 10d	style kNN / XGBoost	0.443	random / Ridge	0.430	Real relation narrowly wins, but remains below the momentum ICIR.

Failed ablations and negative controls. Several implemented variants are deliberately kept in the paper as negative evidence (Table 7). This matters because the search did not simply choose the best-looking row and hide the rest: seed-bagging fixed-graph trees increased turnover and underperformed momentum, and Kaggle/Ubiquant-style date-aggregate features did not repair the short-horizon signal.

Table 7: Negative ablations retained in the research record. These variants are not promoted to the final model because they fail the same hold-out and benchmark tests.

Experiment	Hold-out result	Interpretation
24-member fixed-graph tree seed ensemble	Rank IC 0.0171, ICIR 0.517, Sharpe 0.290, turnover 0.649	Seed-bagging reduces variance in form, but does not beat the low-turnover momentum baseline.
Kaggle/Ubiquant-style date aggregates	Best sector 5d test ICIR 0.569, below the momentum ICIR of 1.016	Cross-sectional date-level aggregates are not enough to stabilize the 5d ML signal.

PM-level robustness audit. The Route B 30-day overlay is the best current research lead, but it is still *not* a production-grade alpha claim. A post-training audit treats the full search as multiple testing, applies CSCV/PBO, neutralizes the final score against available style proxies, and stresses capacity and survivorship. The result is mixed: hold-out economics are better than momentum, and the same-target DSR is encouraging, but the full-grid DSR and PBO still force an exploratory interpretation (Table 8).

Table 8: P0 robustness audit for the Route B 30-day residual momentum overlay. DSR is the probabilistic Sharpe ratio after deflating for the expected best result across the searched configurations; PBO is CSCV probability of backtest overfitting.

Check	Result	Interpretation
Test Sharpe before deflation	0.987	Better than 30d momentum's 0.740, before selection correction.
Bootstrap Sharpe 95% CI	[0.019, 1.796]	Positive but wide.
DSR, same-target trials	0.814	Encouraging within the narrow 30d target search.
DSR, full grid trials	0.003	Fragile if charged for the broader experiment zoo.
CSCV PBO	50.0%	High overfit risk across 28 compact Route B strategies.
Factor-neutral Sharpe	1.006	Economics survive proxy score-neutralization, but attribution is weak.
Fama–MacBeth signal t-stat	0.05	No convincing independent residual-alpha coefficient.
Capacity Sharpe at $100M / $1B	0.948 / 0.864	Capacity is plausible at modest scale; participation rises at $1B.
Survivorship stress-haircut Sharpe	0.837	Sensitivity does not destroy the result, but remains uncorrected.

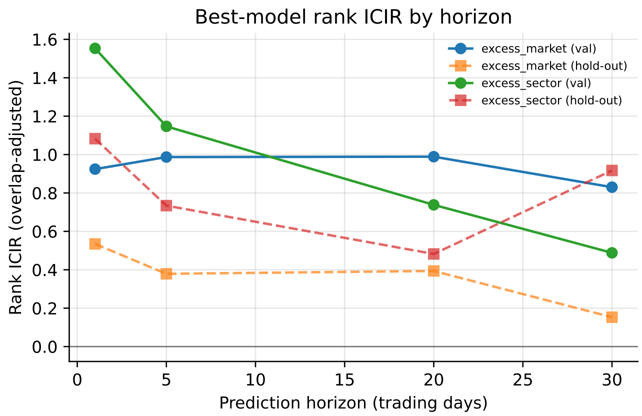

Figure 1: Best-model rank ICIR by horizon (validation vs hold-out), per target family.

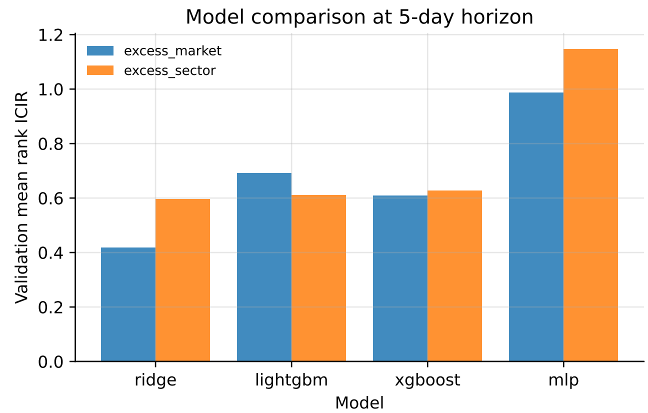

Figure 2: Validation mean ICIR by model at the 5-day horizon.

Overlap correction matters. Fig. 3 contrasts the overlap-adjusted hold-out ICIR with the naive overlapping value. The naive metric inflates sharply with horizon (to ≈3–10 at 30–90 days) while the adjusted metric stays modest (≤ 1.3); reporting the naive number would manufacture a long-horizon "signal" that is pure window overlap. The Sharpe> 3/ICIR> 5 sanity gate flags none of the adjusted runs.

Stability across regimes. Fig. 4 shows per-fold ICIR across the expanding walk-forward. The signal is positive but *variable* across folds, and the hold-out value (dashed) sits below the validation mean — the expected out-of-sample decay. The 90-day horizon is excluded from selection (too few non-overlapping validation periods for a stable ICIR), and the longer-horizon hold-out Sharpe values are estimated from few non-overlapping periods and should be read as noisy. The ensemble search also emits regime slices whenever a split overlaps the default stress windows (GFC 2008, COVID-2020, and inflation-2022); with the fixed ensemble split used here, the available ensemble stress rows are COVID-2020 in validation and inflation-2022 in test, while 2008 is covered only by the broader walk-forward stability analysis.

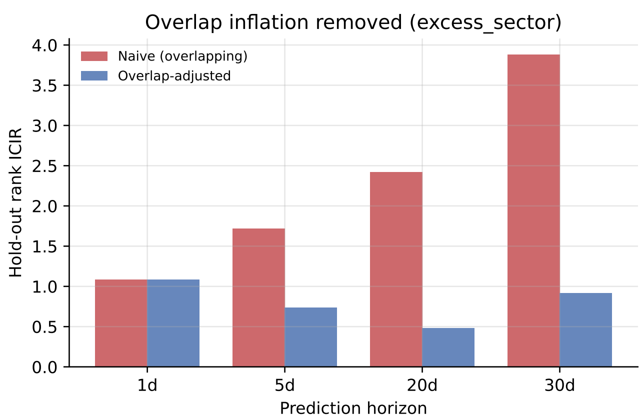

Figure 3: Overlap-adjusted vs naive (overlapping) hold-out ICIR by horizon.

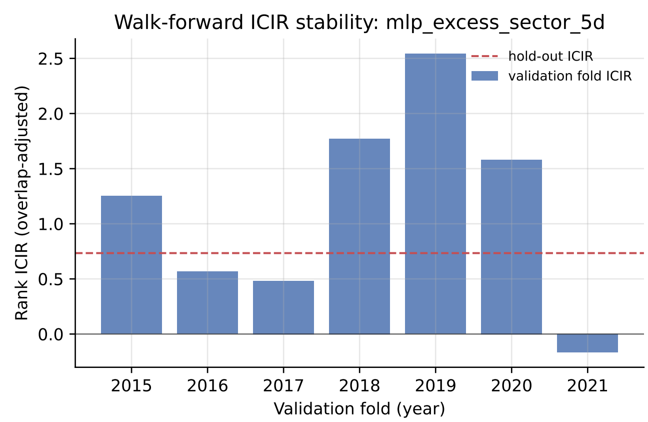

Figure 4: Walk-forward per-fold ICIR for the short-horizon model-comparison run.

Hold-out performance. Table 9 reports the untouched post-2022 hold-out, net of 5 bps/side cost. The numbers are deliberately modest and the simple baseline leads: realistic cross-sectional equity "alpha" from price/volume features alone is small, and the trained models' apparent in-sample edge does not translate into out-of-sample value here.

Table 9: Untouched hold-out (2022–2026), net of 5 bps/side cost. The single-factor momentum baseline beats the validation-selected models on IC, ICIR, Sharpe, return, *and* turnover. Parenthetical ICIR is the naive overlapping value, shown only to document inflation. Best per block in **bold**.

Horizon	Config	Rank IC	ICIR (raw)	Net Sharpe	Ann. Ret	Turnover
5d	Momentum baseline (sector)	**0.024**	**1.06** (2.10)	**1.05**	**13.5%**	**0.16**
5d	MLP *(val-selected, sector)*	0.013	0.73 (1.72)	0.17	1.2%	0.59
5d	Ridge (sector)	0.010	0.40 (0.99)	0.53	5.1%	0.65
5d	LightGBM (sector)	0.009	0.20 (1.17)	0.24	1.7%	0.53
20d	Momentum baseline (market)	**0.043**	0.71 (3.62)	**1.12**	**13.1%**	**0.29**
20d	LightGBM *(val-selected, market)*	0.028	0.39 (2.47)	0.49	5.5%	0.64
20d	Ridge (market)	0.029	0.54 (2.27)	0.76	9.8%	0.61

6 Discussion

Overfitting and the validation–hold-out gap. The clearest evidence of overfitting is in the results themselves: models selected on validation ICIR fail to beat a single momentum factor on the original 5-day raw rank benchmark, and the model with the best validation ICIR (the MLP) is not the best economic hold-out model. The final 30-day Route B rerun improves the momentum book only after the ML component is constrained to a residual sleeve; standalone residual models still have weak economics and high turnover. We use heavy regularization (Ridge penalty, tree depth/leaf/subsample limits, MLP dropout + weight decay + early stopping), select hyperparameters on validation folds only, and report a single untouched hold-out precisely so this gap is *visible* rather than hidden, with the Sharpe>3/ICIR>5 gate catching anything "too good".

Multiple testing. The original grid is large enough that the best Sharpe must be treated as a selected maximum, not as a single pre-specified experiment. For the Route B 30-day overlay, raw hold-out Sharpe is 0.987 and the bootstrap 95% Sharpe interval is $[0.019, 1.796]$. The same-target DSR is 0.814, but full-grid DSR falls to 0.003 and CSCV PBO is 50%. This is the most important statistical conclusion: the overlay is a materially better research lead than the old graph/MLP direction, but still not a statistically deflated production alpha.

Factor attribution. The dataset lacks a full commercial risk model and fundamental value/quality factors, but we can still neutralize the final signal against available style proxies: liquidity/size (`log_dollar_volume`), volatility (`vol_20d`), beta (`beta_60d`), Amihud illiquidity, short-term reversal (`ret_5d`), and 12–1 momentum. The Route B overlay keeps attractive economics under this proxy score-neutralization, but the non-overlapping Fama–MacBeth signal coefficient has only $t = 0.05$. The evidence therefore supports an economically interesting residual sleeve, not a clean claim of independent style-neutral alpha.

Look-ahead leakage. Addressed structurally: one-day execution lag, label purge on both sides, horizon embargo, and train-only preprocessing. We also verified that a no-embargo diagnostic inflates metrics, confirming the controls bind.

Survivorship bias. The dataset is current-membership-on-history, so results are optimistically biased and cannot be treated as a production backtest. The audit adds a numeric sensitivity rather than only a caveat: applying a delisting overlay of annual delisting rate × average delisting loss to the short-side exposure leaves the Route B overlay Sharpe at 0.951 in a base case (2% annual delisting rate, −30% average delisting return) and 0.839 in a stress case (5%, −50%). This bound is still not a true correction; the right fix is point-in-time constituents and delisting returns.

Foundation models. Kronos was tested only as a small zero-shot exploration. On this universe the larger random-date benchmark had negative 2022–2026 test ICs, so it is moved to the "not yet useful" bucket rather than treated as a headline model.

Data quality. Prices are split-adjusted; dividend adjustment is uncertain and treated as a minor caveat. We winsorize features and filter the illiquid decile so bad ticks and microcaps do not dominate.

Turnover, capacity, and market impact. We charge 5 bps per side on realized turnover and rebalance only every h days. Turnover is itself decisive: the trained residual models churn ∼75–85% per rebalance, while the Route B overlay turns over 49.2% versus 40.1% for the 30-day momentum benchmark. A simple square-root impact stress test, calibrated to 10 bps at 1% ADV participation, gives overlay Sharpe 0.948 at $100M AUM, 0.900 at $500M, and 0.864 at $1B, with 95th percentile participation rising to 25.3% at $1B. This is not an execution model, but it is enough to show that capacity and short-side liquidity are first-order constraints.

Regime dependence. Walk-forward folds span 2008, 2020, and 2022 stress periods; Fig. 4 shows the signal is not uniform across regimes, motivating the market/breadth/dispersion regime features rather than a single static model.

Economic intuition. The one signal that clearly survives out of sample is cross-sectional momentum — a well-documented behavioural underreaction effect — which is precisely why the single-factor baseline is so hard to beat. The Route B result is useful because it gives the model a narrower economic job: find residual structure that can be added to, not substituted for, momentum.

7 Limitations and Conclusion

This study delivers an honest, reproducible cross-sectional relative-return pipeline for the S&P 500 using only price/volume and light metadata. The key discipline is methodological: non-overlapping evaluation, walk-forward selection, explicit leakage controls, and a single-factor baseline that every model must clear in a clearly specified metric. The realistic conclusion is sobering but more actionable after the Route B rerun: broad multivariate, graph, and MLP searches do *not* pass a production-grade alpha audit, but a compact 30-day XGBoost residual sleeve improves the momentum book's hold-out Sharpe and drawdown. I would not trade it as a standalone alpha, and I would not promote it without point-in-time constituents, delisting returns, stronger style-factor attribution, and lower-turnover execution tests. The useful output is the triage: keep 12–1 momentum as the core benchmark, treat residual ML as a small research sleeve, and make DSR/PBO, capacity, and turnover part of model selection rather than post-hoc decoration.